TapeUP Bluepaper

From Launch to Build: Native Issuance, Long-Term Market Participation and an Open Building Network

TAPEUP - LAUNCH · BUILD · VERIFY · GROW
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Publication note: This bluepaper is prepared by the TapeUP project team for ecosystem partners, developers, prospective participants and professional reviewers. It presents our mechanism vision, product architecture and development plans. Certain tokenomics, contracts and ITO rights remain subject to final determination and disclosure. This edition is a publication candidate, not a final issuance document that may be used directly for token fundraising or trading promotion.

Conceptual origin: Bruce Lan introduced the Proof of Build (PoB) philosophy. The TapeUP project team designs and develops the PoB Trading Engine and TapeUP product roadmap. This bluepaper distinguishes our existing mechanism designs, proposed implementations and long-term research directions. Feature availability is subject to actual releases.

Core proposition: Token issuance begins the building process; it does not complete delivery. TapeUP aims to connect circuit creation, native asset issuance, capital participation, open development and ongoing services in an evolving Launch & Build Network.

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Executive Summary

Crypto has efficient asset issuance infrastructure, but still lacks infrastructure connecting issuance, actual delivery and long-term collaboration. Tokens acquire a price and liquidity almost immediately, while development, testing, research, maintenance and user adoption take months or years. Rapid attention does not establish a project's ability to keep building.

Using TapeOut's circuit and container architecture, TapeUP designs a product lifecycle anchored in actual delivery and sustained through transparent trading and contribution settlement: DEV project creation and parameter configuration → ITO (transistor minting, circuit construction and tape-out) → automatic deployment of proposed TAP-20 assets under specified rules → PoB Trading Engine → allocation of transaction taxes and protocol revenue → circuit staking and Metro Station services → ongoing development, community building and research collaboration.

TapeUP also plans an Omnichain Ledger: an indexing and verification view that combines multichain data while preserving its original on-chain provenance. It provides traceable source views for cross-chain identities, pages, messages and building evidence. It is not another consensus network. Its purpose is to index, compare and trace records from different sources, showing confirmation status and remaining trust assumptions.

Two complementary concepts must remain distinct. The PoB Trading Engine is the proposed mechanism for scheduled purchases, Builder Codes and points-based rewards. Proof of Build is the longer-term evidence and incentive framework for developer, community and research contributions. The former changes when participation funds enter the market; the latter records real work and supports independent review and transparent funding. Neither substitutes for the other.

TapeUP's long-term objective is to reduce collaboration costs from Launch to Growth through open issuance, verifiable building, sustainable service revenue and governance, rather than guarantee continuous price appreciation. This bluepaper distinguishes existing designs, proposed implementations and long-term research. Statements about prices, returns or delivery dates are not guarantees. Illustrative simulations explain PoB parameter trade-offs: with the same total purchase budget, immediate execution ratios and release periods change the duration of buying, day-one concentration and the shape of a price-path proxy. They do not predict actual market prices.

How to Read This Bluepaper

We distinguish three categories: existing designs (mechanisms or plans we have already proposed), proposed implementations (protocol modules still requiring development, testing and governance approval), and long-term research (optional directions outside near-term delivery commitments). These categories have different maturity levels. Deployment addresses, token rights, security audits and applicable legal scope are subject to subsequent official announcements and contracts.

TapeUP's current product sequence is ITO → native asset issuance → PoB Trading → fee and service settlement → long-term Build. PoB Trading manages capital execution timing; Proof of Build addresses development, research, public goods and community contributions. They support each other but must not be conflated into one set of "building points."

01. Problems, Opportunities and Protocol Positioning

1.1 The Timing Mismatch Between Issuance and Building

Conventional issuance concentrates on token creation, liquidity provision and post-listing trading. Developer collaboration, documentation, performance optimization, security fixes, research replication and market validation take place over longer periods. Market prices move quickly, but cannot establish whether code works, a circuit runs or a service is actually used.

TapeUP therefore addresses three questions: How can a project launch from verifiable delivery? How can early market participation align with long-term building? How can developers, community members and researchers continue to receive evidence-based support after launch?

1.2 Fragmented Multichain Information and Building Evidence

A project may span GitHub code repositories, TapeOut circuits, DeWEB pages, token contracts, on-chain payments, community documentation and cross-chain identities. No single entry point readily establishes which project an item belongs to, what was released, who maintains it or whether it is used. Aggregating data in a database does not establish authenticity. Provenance, blocks, signatures, build records and reviewable context must be retained.

1.3 TapeUP’s Scope

TapeUP is an open protocol and product suite built on existing public chains and TapeOut infrastructure. It does not propose independent L1 consensus or equate cross-chain reading with atomic cross-chain settlement. Its primary value is to connect issuance with delivery, preserve the provenance of building outcomes and make contribution support verifiable, rather than use market price as the sole measure of success.

1.4 Why Start With TapeOut

TapeOut organizes transistor logic, circuit minting, container identities and DeWEB applications into composable objects, giving projects a native identity anchored to a circuit or container. TapeUP can connect ITO delivery events to subsequent asset issuance and operating rules. Circuit identity establishes certain ownership and build records on-chain; it does not automatically establish product quality, commercial demand or organizational credibility.

02. Vision, Principles and Product Architecture

Vision: Build an open Launch & Build Network where projects begin with circuit creation, issue assets, organize long-term participation and give development, maintenance, community and research outcomes a verifiable basis for recognition and settlement.

Mission: Lower issuance barriers; improve asset and contract transparency; extend the building window; establish cross-project contribution records; and cultivate genuine demand for services.

2.1 Six Design Principles

- **Build before Narrative:** Prioritize reviewable deliverables over marketing promises.

- **Rules before Trading:** Disclose economic parameters, permissions, fund destinations and exit rules before accepting funds.
- **Verifiability by Default:** Prioritize source-chain facts, public code, signatures and independent recomputation; introduce more complex cryptography only when needed.
- **Separation of Ledgers:** Record capital participation, software development, research, community work and actual revenue separately to prevent conflated scoring.
- **Open Contribution:** Building extends beyond the founding team. External developers, maintainers, researchers and community members can contribute through defined tasks.
- **Progressive Decentralization:** Decentralize permissions, upgrades, custody and service provision over time, disclosing the centralization risks remaining at each stage.

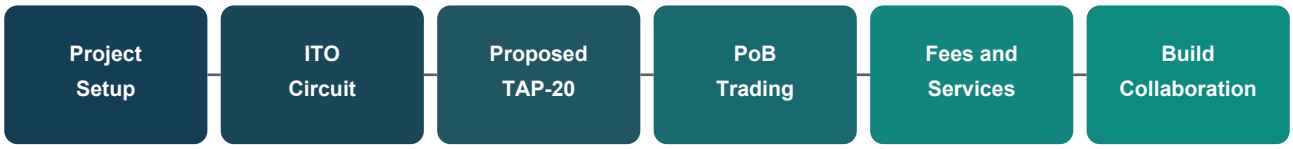
2.2 Product Stack

Layer	Core Components	Primary Function	Stage
Information and identity	Omnichain Ledger, chain adapters, verification clients	Verify source-chain records and cross-chain identities	A
Project issuance	Launch Factory, ITO Gateway, asset configuration	Trigger standardized issuance from circuit creation	A process, B implementation
Assets and compatibility	Proposed TAP-20, asset-state module, compatibility interfaces	Supply, transfers and asset discoverability	A proposal, B implementation
Market participation	PoB Trading Engine, Vault, Scheduler	Immediate and scheduled purchase execution	A mechanism, B implementation
Service operations	Circuit staking, Metro Stations, task verification	Settle against verifiable service records	A plan
Long-term building	Build Ledger, Build Treasury, research and community tasks	Support actual long-term building	B → C
Optional verification	Private eligibility proofs, specific computation proofs	Stronger verification where required	C

Architecture: The base layer supplies on-chain provenance and circuit identity; Launch Factory creates project instances; the asset layer defines rights and supply; the PoB engine manages purchase budgets only as authorized; Metro Stations provide verification services; Build Ledger records long-term contributions; and Build Treasury pays contribution rewards within governance-approved budgets.

03. TapeUP Launch Lifecycle From DEV to Ongoing Operations

TapeUP’s launch lifecycle starts with circuit creation and verifiable delivery, connecting asset issuance, market participation and ongoing operations. The following state machine describes our proposed process and state boundaries. Parameters not yet determined will be disclosed before implementation and release.



Provenance, building evidence and settlement records span the lifecycle

TapeUP six-stage lifecycle overview

3.1 Step 1 - DEV Project Creation and Parameter Commitments

Developers submit the project name, website and repository, chain and circuit information, target users, issued asset description, supply rules, ITO success criteria, initial allocation principles, PoB parameters, transaction tax collection method and receiving addresses. The protocol generates a unique projectId and versioned configHash.

Proposed commitment mechanism: Before accepting user funds, publish key rules as a hash commitment, human-readable configuration and on-chain events. Parameters affecting economic rights must not be changed retroactively without authorization. Upgradable fields, upgrade administrators and emergency pause conditions must also be shown.

3.2 Step 2 - ITO Transistors, Construction and Tape-Out

Our planned ITO process includes transistor minting, circuit design and construction, tape-out completion and satisfaction of predefined conditions. Each issuing project should link a specific circuit identity, minting or construction transactions, required material records, result hashes and holder control relationships.

Scope of the term: In this document, ITO means circuit creation/minting and related qualification events within TapeOut. It does not mean physical wafer fabrication in the conventional semiconductor industry, nor imply that every ITO produces a physical chip tested by a third party. Completion depends on verifiable states in the relevant factory and application contracts.

ITO participant rights remain to be determined. Circuit NFTs, token allocation eligibility, automatically allocated tokens and combinations of these rights must be defined in each issuance scheme. Before participation opens, we will disclose the rights, acquisition conditions and risks. Undetermined rights will not be described as “automatic returns.”

3.3 Step 3 - Automatic Issuance of Proposed TAP-20 Assets

Launch Factory initializes assets only after predefined ITO completion conditions are verified, configuration commitments match deployment permissions, and replay-protection and uniqueness checks pass. Issuance fixes or discloses the asset identifier, initial supply, supply cap, allocation addresses, administrative permissions and compatibility approach.

Technical prerequisite: TAP-20 is TapeUP’s proposed specification application and has not been formally adopted through TapeOut’s standards process. Automatic TAP-20 deployment is therefore a target state. If initial launches use ERC-20 or another compatible asset, the interface will state the current asset type and any future migration conditions. Integration with existing DEXs still requires appropriate adaptation and testing.

3.4 Step 4 - Launch of the PoB Trading Model

Before issuance, each project selects and discloses PoB parameters, including the immediate execution ratio, future execution period and transaction tax rules. Users voluntarily place a project-specific PoB order, authorizing immediate and scheduled budgets. Each order links to the project instance, purchased asset, DEX routing rules, minimum execution requirements and fee breakdown.

Important distinction: PoB manages buyer funds; it does not impose a mandatory token lock on all holders. Funds pending execution are unfilled purchase budgets, not tokens already acquired.

3.5 Step 5 - Transaction Taxes, Buybacks and Fund Allocation

Project transaction taxes, collected on specified trades under disclosed rules, and protocol service revenue must be recorded and disclosed separately by source, then routed to the relevant accounts under preannounced allocation rules. Escrowed purchase principal must never be treated as transaction tax revenue or building-fund income. Protocol buybacks, operating costs, ecosystem funding and safety reserves require explicit allocation ratios and permissions.

3.6 Step 6 - Circuit Staking and Metro Station Services

Eligible circuit holders may apply for staking and Metro Station services. Eligibility may depend on circuit ownership and staking requirements, but service compensation must relate to valid reads, verification, availability or independently recognized work. Reward funding, exit procedures and dispute rules must be determined before real funds are accepted.

3.7 Step 7 - Long-Term Build as Ongoing Practice

A project is not complete when its token is issued. TapeUP aims to maintain records, tasks, reviews and budget support for development, research, community maintenance and actual usage. This is the long-term Proof of Build framework, not a renaming of existing PoB trading points.

3.8 Lifecycle State Boundaries

State	Trigger	Public View	Fund Risk Controls
Draft	Developer creates configuration	Undeployed and editable	No public funds accepted
Committed	Public commitment to key parameters	Configuration version and hash	Reconfirmation for changes
ITOActive	Minting/construction begins	Contracts, quantities, completion rules	Fund and asset ownership disclosure
ITOComplete	Verifiable conditions met	Completion evidence and finality	Prevent duplicate triggers
AssetInitialized	Successful issuance	Asset ID, supply, control	Invariant checks
MarketActive	Venue and PoB ready	Liquidity, slippage, orders	Custody and order exit rules
Operating	Ongoing products and services	Building, usage, settlement	Budget/staking/permission audits
Disputed / Paused	Anomaly or material dispute	Reasons, remedies, timeline	Pause only under predefined powers

04. ITO and Fair Issuance Design

4.1 ITO Beyond a Token Issuance Gateway

A meaningful ITO requires an identifiable issuer, an inspectable circuit or container deliverable, predetermined issuance conditions, participant understanding of acquired rights, and disclosure of unfinished R&D and commercial risks by the issuing project. Calling an ordinary token issuance an ITO without circuit delivery constraints does not create differentiation.

4.2 Four Issuance Models for Consideration

Model	Participant Rights	Suitable Context	Risks
Circuit-first	Circuit assets/container access eligibility	Developer-oriented building projects	Uncertain NFT value and delivery acceptance
Asset-first	Token allocation eligibility under explicit rules	Community token issuance	Securities/fundraising issues and allocation fairness
Combined	Separately defined circuit and token rights	Infrastructure and application ecosystems	Distinguish dual rights and valuations
Milestone-based	Initial eligibility; designated incentives released against results	Long-term research/ecosystem building	Review disputes, custody and governance

Proposed configuration principle: First define a default Profile, then gradually introduce optional templates. Each template must have distinguishable risk disclosures and contract parameters that issuing projects and participants can verify.

4.3 Distribution and Abuse Prevention

Options include wallet holding requirements, early-contributor eligibility, public lotteries, participation caps, commit-reveal subscriptions and allowlist proofs. Each involves trade-offs in Sybil resistance, information asymmetry and operational complexity. No mechanism should be described as inherently “fully fair.” ZK-based anonymous membership may be studied as a future privacy tool, but is not a technical dependency of the first ITO release.

4.4 Issuance Readiness Checks

Before issuance, complete build attestations, asset-state tests, a fund ownership map, lockup and release policies, a liquidity plan, an administrator permission list, risk disclosures, exit mechanisms, a testnet demonstration and independent security review. Product pages should show whether each check passes rather than present an ambiguous “safety score.”

05. Proposed TAP-20 Asset State and Ecosystem Compatibility

5.1 Current Proposal Status

As of this edition’s preparation date, TapeUP had discussed circuit/container-native assets through TapeOutProtocol/TAPs Issue #21 and submitted PR #23 to the official repository. That PR was closed / not merged. The application name “TAP-20” had not received formal editorial numbering or adoption; the proposal provisionally identifies its specification as tape-20/v1. Our public materials will disclose this status and will not present it as an adopted standard.

5.2 Technical Features of the Proposal

The proposal seeks to bind asset issuance to TapeOut circuit identity, producing replayable records through deploy / mint / transfer / burn operations. It explores a specification-governed runtime for total supply, balances and transaction ordering. This differs from defining all balances directly through an ordinary balanceOf mapping in a standalone ERC-20 contract. The implementation must still identify the final authoritative source of asset state.

5.3 Asset Invariants and Execution Principles

Minimum invariants: 0 ≤ totalSupply ≤ supplyCap; aggregate balances equal total supply; transfers do not change total supply; minting and burning follow initial rules; failed operations neither modify balances nor consume a nonce. Signatures must bind chainId, runtime contract, asset ID, recipient, amount and expiry to prevent replay. Events may support indexing, but their existence alone does not establish a consistent, adjudicable final asset state across clients.

5.4 ERC-20 and DEX Interoperability

Existing wallets and DEXs rely on ERC-20 balance queries, transfer, allowances, transferFrom and standard events. Native assets lacking these capabilities require reviewed adapters or wrapping mechanisms, with explicit underlying-asset locking, wrapped supply conservation, redemption, pause and custody rules. Compatibility must be tested end to end, not inferred from an asset’s name.

5.5 Phased Rollout Strategy

We propose separating issuance orchestration from the asset standard. Launch Factory retains a common AssetAdapter and selects eligible implementations by public version. Proposed TAP-20 support would be introduced as test vectors, compatibility and the external standards process mature. Any migration must define conversion responsibilities, timing and failure recovery. User assets must not be migrated unilaterally without authorization.

06. PoB Trading Engine An Order Protocol for Long-Term Participation

6.1 Mechanism Parameters

The PoB Trading Engine offers three configurable parameters: an immediate purchase ratio of 10%-90% in 10-percentage-point increments; a scheduled buying release period of 1-365 days; and a transaction tax of 1%-10%. Charging order, applicable tax base and network fees still require specification in the contracts.

Each buy order comprises an immediate purchase budget and an unfilled future purchase budget. The latter becomes executable linearly over a predefined period. Overlapping orders create batches with different maturities. The mechanism changes execution timing, not purchase principal, and does not guarantee price appreciation.

6.2 Order Accounting and Formulas

Let the user’s gross payment be P, all lawfully and contractually deductible fees be F, the net executable budget be N=P−F, and the immediate ratio be α:

ImmediateBudget = α × N
ScheduledBudget = (1 − α) × N
BaseCumulative(t) = ScheduledBudget × clamp(t / T, 0, 1)
CumulativeExecuted(t) ≤ ScheduledBudget + any unspent immediate remainder
EscrowLiabilities = Sum(unspent order balances + refundable balances)

Acceleration and failures mean that actual execution cannot be determined solely from BaseCumulative(t). The proposed implementation separately records each order’s cumulative authorized executable amount, cumulative successfully executed amount, cumulative refunds and available unexecuted balance, using fixed-precision arithmetic and idempotent order identifiers.

Example: For 1,000 USDT with 30% purchased immediately and 70% executed linearly over 25 days, excluding taxes, Gas, slippage and failures, 300 USDT is purchased immediately and 700 USDT awaits execution. Without acceleration, the daily planned budget is 28 USDT. Tokens received depend on actual execution prices.

6.3 Builder Code Bilateral Acceleration

Each participant has a Builder Code. A valid Code may be entered once per order, including separately on multiple orders. Participant A advances execution of 20% of A’s pending-release amount; Code owner B simultaneously advances B’s own balance by no more than that amount. A Code is invalid if B has no remaining execution budget. Accelerated funds must still buy the same project’s token and be deducted from each participant’s subsequent schedule.

Example: If A's applicable balance is 700 USDT, A accelerates 140 USDT. B accelerates 140 USDT if B has 500 USDT remaining, or 80 USDT if only 80 USDT remains. Final rules must determine whether the 20% base is per order or account-wide, along with execution order, trigger limits and failure rollback.

6.4 Points and Daily $UP Buyback Distribution

PoB records actual accelerated amounts and distributes protocol-bought $UP in proportion to newly accrued valid points for the day. If distributable tokens on day d are Qd and participant points are si,d, an illustrative allocation is:

Reward(i,d) = Qd × si,d / Sum(sj,d)
Condition: total valid points > 0. Otherwise, no proportional distribution occurs that day; the disclosed fallback rule applies.

These points record market collaboration and acceleration, not GitHub commits, research, documentation or maintenance. Points should become valid only when the accelerated purchase actually executes. Reciprocal Code farming, self-trading, repeated acceleration across related accounts and abnormal funding sources require review.

6.5 User Funds, Cancellation and Execution Risks

Contracts must specify who controls the receiving Vault; withdrawal or cancellation rights before execution; fee deduction timing; retries for insufficient liquidity or excessive slippage; principal protection during pauses; remedies for malicious DEX route replacement; mitigation of predictable MEV on public scheduled orders; and orderly withdrawal of remaining funds when a project terminates.

Core accounting identity: net user deposits always equal executed purchase budgets + unexecuted budgets still in custody + refunds. Fees must be separately disclosed and reconciled item by item. Pending budgets cannot be recognized as protocol revenue or counted twice when release schedules change.

6.6 Practical Boundaries of the PoB Trading Model

6.7 PoB Curves and a 180-Day Scenario Experiment

We define four hypothetical models to compare funding schedules, not forecast prices. Every model has a purchase budget of 100 units. Only the immediate execution ratio and the remaining execution period differ.

Model	Immediate	Scheduled Execution	By Day 7	By Day 30	Execution Days
Immediate	100%	None	100%	100%	1 day
Fast PoB	60%	Evenly over 14 days	80%	100%	15 days
Balanced PoB	30%	Evenly over 25 days	49.6%	100%	26 days
Long-tail PoB	20%	Evenly over 90 days	26.2%	46.7%	91 days

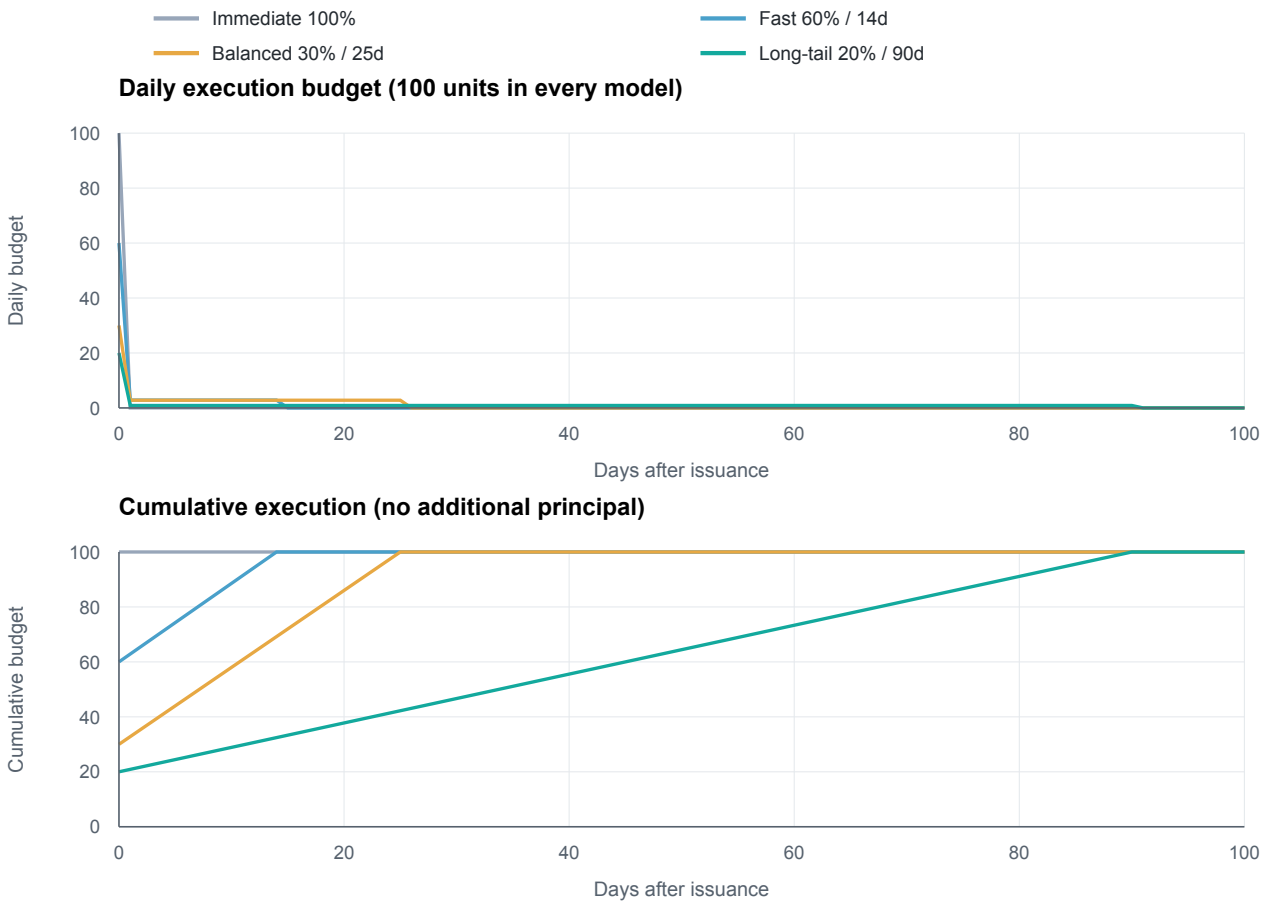


Figure 1 | Daily and cumulative execution for four entry models with equal total budgets. Acceleration, fees, failed trades and refunds are excluded.

The chart shows PoB's directly testable structural difference: a longer execution window and less concentration on day one, without increasing the total purchase budget. "Sustained buying" means some orders retain restricted future execution budgets; it does not mean that the protocol creates new funds.

6.7.1 Two External Market Scenarios

We compare sustained post-issuance selling pressure with early weakness followed by improving demand. Within each scenario, the four strategies face identical external market changes and differ only in buy execution timing. The index uses an artificially specified linear log-impact rule, not historical BEM, $UP or other real token data.

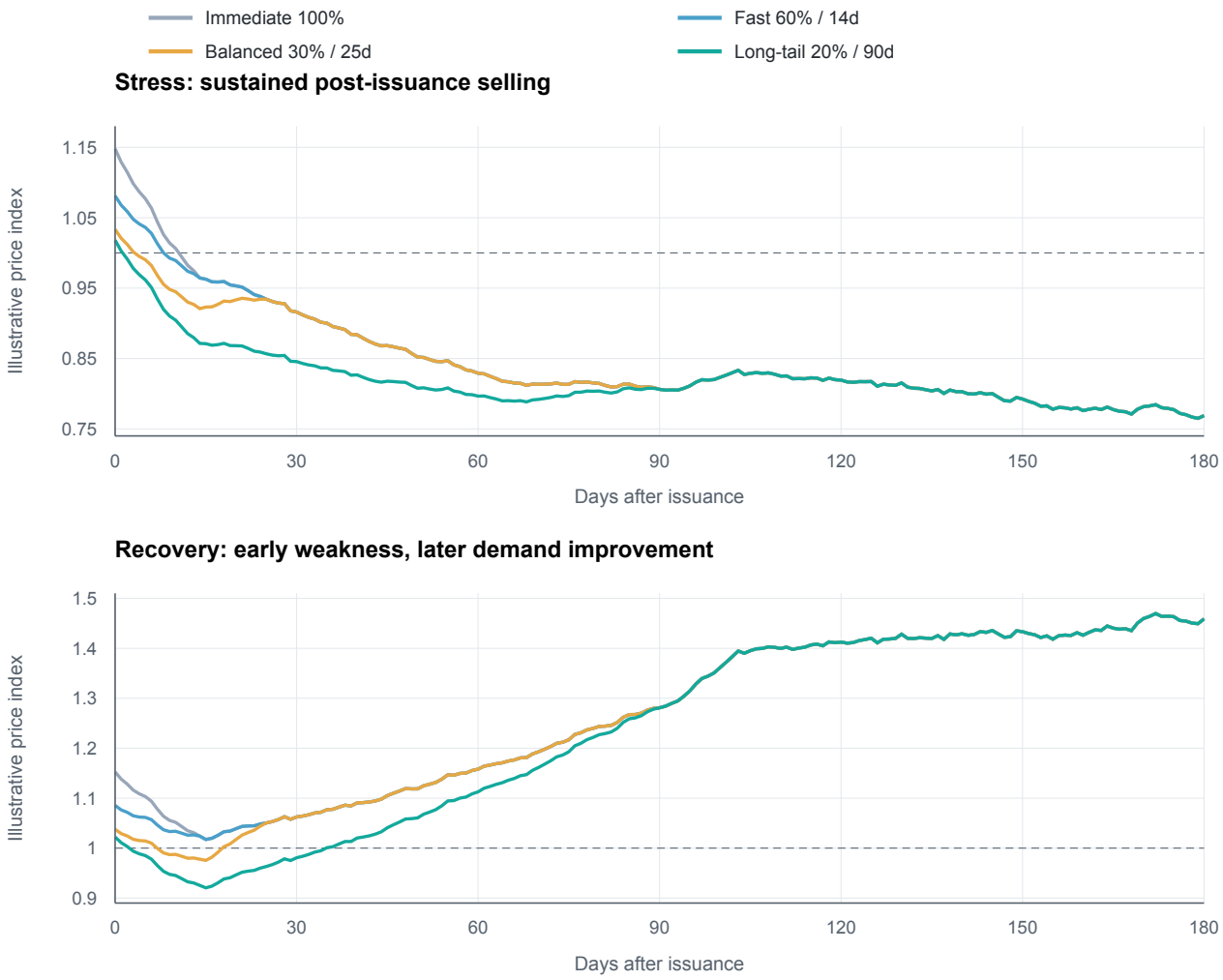


Figure 2 | Normalized price-path indices in two hypothetical markets. These indices are neither AMM quotes nor price forecasts.

The stress scenario shows that persistent external selling can keep the simulated index below the issuance baseline even with long-tail buying. The recovery scenario shows that long-tail execution may reduce early impact but can also face a longer weak period before underlying demand recovers. Neither result demonstrates that a template is safer or necessarily yields higher returns.

In this simplified linear construction, where all budgets execute successfully, cumulative purchases are identical after day 90 and indices coincide at day 180. This is a mathematical property of the model, not necessarily a result in real AMMs with arbitrage, fees, slippage, changing liquidity and strategic behavior.

6.7.2 Model Selection and Parameter Trade-Offs

Dimension	Fast	Balanced	Long-tail
Early liquidity impact	Relatively strong	Moderate	Relatively weak
Participation window	Short	Medium	Long
Reliance on later executors and scheduling	Lower	Moderate	Higher
Cost risk of frequent small executions	Medium	Medium	High; depends on thresholds and aggregation
Research use case	Rapid price discovery	Launch-stage balance	Long-cycle delivery experiments

These configurations are research templates, not approved default trading tiers or live settings. The configurable ranges remain 10%-90% and 1-365 days, as stated in Section 6.1.

6.8 Simulation Method and Applicability

Daily scheduled budgets are buy_m(t) = 100×α on day 0 and buy_m(t) = 100×(1−α)/H on days 1 through H, and zero thereafter. The simulated index is I_m(t)=exp(Σ[g_s+ε_s+κ×buy_m(s)]), where g_s is the assumed daily external market change, ε_s is the disturbance sequence shared by all four models, and κ=0.0015 is an arbitrarily chosen impact coefficient. Figure 2 does not model full AMM reserves, order books, cross-market arbitrage or real token price formation.

Appendix D provides the piecewise assumptions and reproducibility details. After launch, PoB's actual effects should be tested against real orders, executed volume, retention, slippage, liquidity and project delivery. It can create more time for building, but cannot prevent price declines or replace genuine product demand.

07. Protocol Fees and Fund Flows Keeping Three Fund Categories Separate

TapeUP's economic structure contains at least three separate categories: user purchase budgets, recognizable protocol revenue, and dedicated building and node-reward budgets. Ownership, risk, use and audit treatment must remain separate.

7.1 The Role of Transaction Taxes

PoB's transaction tax is configurable from 1% to 10%. Taxed transaction types, deduction order, receiving addresses, modification permissions and cross-chain allocation rules remain to be finalized and disclosed before each project opens participation. This range does not imply a uniform rate across instances, and the maximum is not the default. Project pages must show actual parameters and the effective after-tax execution amount.

7.2 Proposed Fund Segregation

Account	Source	Uses and Payment Constraints
Order Vault	User budgets for their own orders	Execute orders or refund under disclosed rules only
Protocol Revenue	Disclosed protocol and service fees, lawful transaction tax shares	Operations, security, ecosystem budgets and potential buybacks
Build Treasury	Public ecosystem allocations, grants, other funding	Development, research, maintenance and community tasks after acceptance
Service Settlement Pool	Actual Metro Station customer payments	Settlement for valid services under SLA and dispute rules
Safety Reserve	Explicit risk provisions	Audits, emergencies and governance-approved security spending

Projects may select appropriate subaccounts, but every fund must have a traceable source, beneficiary, spending authority and current liabilities.

7.3 $UP Economic Status Pending Formal Disclosure

We use $UP as the identifier for the proposed asset distributed through daily protocol buyback rewards. As of this edition's preparation date, total issuance, allocation, vesting, contract addresses, rights, circulation plans and issuance status had not been fully disclosed. Formal issuance documents and announcements will govern.

This bluepaper treats $UP as a proposed protocol reward asset pending formal disclosure. If later designs include payment, governance or staking functions, we will separately explain acquisition, rights and risks. These optional functions are not presented as effective rights in this edition.

7.4 Budget Constraints on Buybacks and Building Funds

For a period with realized net protocol revenue R, execution costs C_exec, operating costs C_ops, safety reserves C_risk, building grants G and a buyback budget B:

$$B + G \le \max(R - C_{exec} - C_{ops} - C_{risk}, 0) + PubliclyApprovedSubsidy$$

This formula expresses budget discipline, not a return forecast. Any initial external subsidy must disclose its source, amount, duration and exit conditions. Buyback budgets, unexecuted user principal and token market capitalization cannot be double-counted as protocol revenue.

7.5 Token Rights and Transparent Disclosure

TapeUP distinguishes protocol assets, assets issued by individual projects, the proposed $UP reward asset and circuit NFTs. Payment, access, staking, governance and revenue-sharing rights depend on final agreements, smart contracts and applicable law for each asset. As of this edition, we had not fully disclosed $UP supply, allocation, vesting, deployment addresses or specific holder rights. Undetermined tokenomic parameters are not presented here.

Formal issuance disclosures should identify asset functions and limitations, minting or burning powers, allocation and vesting, administrator permissions, revenue-sharing rights, territorial restrictions and any operating commitments that contracts cannot guarantee.

08. Circuit Staking and Metro Stations From Eligibility to Actual Service

8.1 Planned Participation Path

Our planned path is transistor materials → circuit design and minting → staking designated circuits → Metro Station reading and verification → contribution-based service compensation. Metro Stations read source-chain records, verify page fingerprints and message digests, and supply structured data to the Omnichain Ledger.

8.2 Node Service Tasks

The proposed task structure includes taskId, sourceChain, blockRange, inputHash, requester, deadline, serviceLevel, outputHash, nodeSignature and validationResult. Independent validators can recompute replayable tasks using identical inputs and compare outputs. Non-replayable tasks must identify their evidentiary limitations.

8.3 The Economics of Staking

Staking can create economic constraints and an entry threshold, but does not automatically prove correct data, node independence or completed work. We propose basing node compensation on verifiable tasks, quality, availability and actual customer payments, not a simple linear function of staked circuit count. Slashing, transfers of staked NFTs, exit waiting periods and appeals still require clear rules before services open.

8.4 Revenue Sustainability

The long-term priority is actual payments by wallets, DeWEB browsers, project clients and developer services for reliable data or verification. Token incentives may provide transparent temporary subsidies, but cannot replace demand. Node revenue should be settled after on-chain fees, indexing, storage, bandwidth, independent verification and safety reserves.

09. Omnichain Ledger A Provenance Verification Layer for Open Data

9.1 Positioning and Initial Release Goals

TapeUP’s Omnichain Ledger provides a unified reading view for identities, pages and messages while retaining source-chain provenance. Clients verify records under public rules. We previously targeted a limited public preview by the end of October 2026. This is a planned milestone; actual availability and launch status depend on official releases.

9.2 Data and State Model

Proposed record fields include chainId, source contract address, object ID, block height and hash, transaction or event locator, original payload hash, retrieval time, confirmation status and verification method. Clients must distinguish confirmed, pending finality, conflicting provenance, unavailable data and unverifiable records. Chain-specific finality policies are maintained publicly through versioned configurations.

9.3 Cross-Chain Does Not Mean Trustless

Cross-chain indexing improves usability but cannot eliminate correlated RPC failures, chain reorganizations, bridge risks or third-party node collusion. Metro Stations should publish node independence assumptions, conflicting records and recovery methods. Cached indices do not replace final source-chain state. Asset transfers and cross-chain settlement, in particular, cannot receive strong security assurances solely from an aggregator's response.

9.4 Relationship to Proof of Build

As an evidence gateway, the Omnichain Ledger can display circuits, deployments, on-chain funds, visible code versions and service tasks. It does not unilaterally certify project quality or research validity. Evidence presentation should remain separate from quality judgments.

10. Proof of Build Long-Term Building Beyond Issuance

10.1 From Building Obligations to Open Collaboration

Proof of Build starts from a timing problem: price discovery happens too early and building happens too late. A project receives capital, attention and initial participants at issuance, creating an expectation of continued delivery to its community. Market participation can make time for building, but cannot replace it.

TapeUP proposes extending PoB beyond early market participation into a long-term collaboration framework for Development (development and maintenance), Community (community work and public goods), and Research (research and experimental replication). These contributions require different evidence and evaluation methods and should not be collapsed into one easily manipulated score.

10.2 Four Ledgers as a Common Language for Building

Ledger	Key Question	Example Evidence	Common Misinterpretation
Build	What works and can be reproduced?	Releases, build hashes, circuit IDs, tests, code reviews	Treating commit counts as quality
Usage	How many independent users actually use it?	Deduplicated calls, retention, real external dependencies, payments	Counting bot calls as growth
Settlement	Where does money come from and go?	Payment proofs, grants, fee accounts, reward details	Treating trading volume as revenue
Market	How do funds and liquidity operate?	Vault balances, executions, slippage, liquidity	Treating scheduled buying as completed purchases

10.3 Start With Evidence Rather Than an Aggregate Score

We propose first providing a verifiable fact dashboard and task-specific acceptance results. A critical security fix, a widely reused circuit component or a reproducible research report may be more valuable than many ineffective commits. If contribution reputation is introduced, we will prioritize contextual, explainable, traceable and appealable records, rather than force every activity into one number.

10.4 Keep Two Incentive Systems Separate

Dimension	PoB Trading Participation	Proof of Build Contribution
Measured activity	Orders, accelerated executions, participation rules	Deliverables, maintenance, research and community work
Direct evidence	Execution hashes, balances, Code triggers	PRs, releases, tests, research, acceptance, service outcomes
Typical rewards	Buyback $UP under the mechanism (proposed)	Grants, bounties, project funding or contribution rewards (proposed)
Core risks	Wash volume, reciprocal farming, MEV, trading losses	Poor delivery, attribution disputes, review rent-seeking
Accounting	Recomputable trading records	Category-specific evidence and independent acceptance

Both incentive systems may share an ecosystem interface, but similar naming does not justify sharing unvalidated point weights.

11. Build Ledger Recording GitHub, Code and Circuit Contributions

11.1 A Building Record for Developers

Projects may link one or more GitHub repositories, circuit IDs, DeWEB deployments and public releases to a continuously updated Build Ledger. Repository links may use administrator authorization, GitHub Apps or signed statements, recording validity periods, provenance and revocation history.

Public data is not inherently trustworthy. A merged GitHub PR proves that a repository recorded a merge; it does not prove secure code, independent contributors or production usage.

11.2 Recordable Evidence

Contribution	Primary Records	Stronger Acceptance Evidence
Core feature development	PRs, changesets, linked issues	Merge + CI + release + requirements acceptance
Bug/security fixes	Issues, corrective PRs	Reproduction and regression tests, risk review
Circuit optimization	Circuit/container IDs, build files	Output equivalence, reproducible experiments and metrics
Open-source reuse	Dependencies, third-party integrations	Independent adoption, production deployment, citations
Routine maintenance	Releases, continued adaptation	Response time, stability, review quality
Documentation and developer tools	Tutorials, SDKs, examples	Independent developers reproduce the documented steps

GitHub Webhooks require signature checks and audit trails for deletion, rewriting, force-pushes and repository migration. High-value outcomes should record commit SHA, release tags, build-tool versions and content hashes. Revoking identity links must not erase historical facts, but attribution labels should be corrected.

11.3 Do Not Reward Low-Quality Commit Counts

Our building incentives do not pay directly for Commit or Issue counts. Tasks should define the problem first, then allocate support according to delivery, peer review, replication and adoption. Related accounts submitting large volumes of low-value changes require anti-abuse review.

11.4 Respect Existing Open-Source Practices

Build Ledger must not place repositories under exclusive control or strip authors of copyright or licensing rights merely because their work is recorded. Funding must respect repository and dependency licenses, contribution agreements and research attribution practices. Without authorization, private code may be represented only by necessary metadata commitments, not published without consent.

12. Community Track Community Beyond Attention Farming

12.1 The Work of Community Building

Sustained open-source projects rely on documentation, translation, testing, developer support, technical training, tool maintenance, cross-project collaboration and governance discussions. These contributions can lower entry barriers just as code does. We propose formal, verifiable participation paths for non-programmers.

12.2 Proposed Task-Based Incentives

Projects or self-governing community groups publish task descriptions, deadlines, reviewers and budgets. Contributors submit reviewable work, followed by acceptance and payment records. Ongoing operations can use periodic reviews of response quality, problem resolution and independent feedback, rather than counts of posts or invitations.

12.3 Community Governance and Abuse Prevention

Promotional reach must be assessed separately from genuine developer conversion and retention. Referral commissions should not be the sole community economic model. Rules should prohibit duplicate content, bot reposts, fictitious workshops, reciprocal reviews by related accounts and bad-faith reservation of tasks. Appeals, re-review and evidence updates should prevent one administrator from permanently defining a contributor’s reputation.

13. Research Track Open Research as Building

13.1 Why a Research Track

TapeOut involves frontier topics in circuits, compilation, reusable logic gates, computation models, verifiable computation and system performance. Research may resist immediate product KPI measurement yet generate lasting reusable value. Our proposed Research Track provides clear task channels and visible records for exploratory and replication work.

13.2 Research Funding Types

Type	Required Deliverables	Review Focus
Exploratory Research Grant	Problem, methods, progress, interim results	Research design and execution credibility
Open Bounty	Explicit, testable technical objective	Valid solution and reproducible performance
Replication Grant	Code, data, environment, rerun scripts	Agreement with original results and explanation of differences
Benchmark Prize	Public dataset, evaluation scripts, baselines	Comparability, statistics, reproducibility
Negative Results	Complete failed experiments and analysis	Methodological rigor and public knowledge value

13.3 Evaluate Research Rather Than Paper Packaging

Public papers or preprints, DOIs, test vectors, experiment logs, code and open-source licenses may serve as evidence. Citation counts, social attention and speaking appearances do not replace research quality. Key experiments should permit independent replication. Non-public commercial data may require summaries, third-party attestation and limited-scope disclosure.

13.4 Research to Build

When research becomes circuit optimization, an open-source library, a standards proposal or an application, Build Ledger may link citations and reuse with author permission. Citations do not automatically create cash entitlements. The graph supports transparent attribution, future grant decisions and upstream maintenance funding, not mandatory fees on open-source dependencies.

14. Build Treasury Coordinating Sustained Building Funds

14.1 Funding Sources and Principles

Build Treasury may receive preannounced ecosystem allocations, governance-approved protocol revenue budgets, sponsorships or grants. User PoB principal does not belong to it, and customer payments for node services must not be diverted automatically. Building funds require independently verifiable balances, disbursement records and outstanding commitments.

14.2 Four-Stage Task Process

Proposal → Budget Approval → Delivery & Review → Settlement. Each task should identify its contribution category, reviewers, budget currency, milestones, acceptance evidence, intellectual property requirements, dispute period and payment triggers. Complex research permits reasonable staged funding; not all research funding should depend on a successful final outcome.

14.3 Milestone-Based Incentive Release

Projects may voluntarily use Build-to-Unlock to release designated ecosystem grants against code releases, independent security reviews, successful deployment, real usage or research replication. Use contract facts for objective on-chain events. Subjective quality assessments require public review, appeals and delayed execution, rather than claims that oracle code can “automatically prove everything.”

14.4 The Actual Constraints on Sustainability

Each budget cycle should disclose realized revenue, approved spending, unpaid commitments and net balances. Ecosystem funds should neither depend indefinitely on transaction taxes nor promise an unfunded fixed APR. Grants are budget allocations to public goods, not dividend rights inherently owed to token holders.

15. Builder Passport and Contribution Graph

Long-term research may explore a portable Builder Passport that records voluntarily linked achievements across projects for developers, researchers and community members. It is not a real-name identity system, does not force wallets to be linked to one individual and does not permanently define personal worth through a reputation score.

Contribution Graph can trace sourced relationships in circuit reuse, software dependencies, standards citations, research replication and maintenance collaboration. Upstream components reused by independent projects may merit more funding, but download counts, forks or automatically counted dependencies should not alone determine allocations.

We propose explainable evidence links first. More complex reputation and public-goods allocation models can be researched as Sybil resistance, identity authorization, correction and governance improve.

16. Verification Roadmap ZK as a Tool Rather Than the Central Narrative

TapeUP proposes this verification priority: deterministic on-chain facts → public source code and reproducible builds → signatures and third-party review → challengeable dispute mechanisms → cryptographic enhancements where needed.

ZK is a long-term research direction for anonymous eligibility, private-condition verification, correctness of complex computation and selective disclosure of aggregates. It is not an MVP prerequisite. We consider it only where signatures, on-chain events and public data are insufficient and proof costs are acceptable.

Fundamental limitation: ZK proves a computational relationship for given inputs and rules. If a project supplies fabricated inputs, ZK does not turn them into real commercial usage. NAND/LATCH circuits and ZK constraint systems are not directly interchangeable. Any ZK release should disclose its proof system, trusted setup or recursive assumptions, verification contracts, costs and audit status.

17. Security Architecture and Market Integrity

17.1 Smart Contract Security

Fund-handling modules - Launch Factory, asset Runtime/Adapter, PoB Vault, Execution Scheduler, Fee Splitter, Staking and Treasury - should each undergo access-control analysis, threat modeling, unit tests, fuzzing, state-invariant tests, independent audits and capped private testing. Key risks include reentrancy, incorrect authorization, duplicate accounting, invalid asset mappings, malicious DEXs, time manipulation and upgrades.

17.2 Market and Execution Security

Public scheduled purchases expose predictable execution, MEV front-running, price impact and thin-pool arbitrage risks. Acceleration creates risks of related-account loops, deliberate liquidity capture and fabricated points. Limit prices, slippage caps, variable execution windows, batch matching and protected routes may be studied, with costs, availability and centralization dependencies assessed for each.

17.3 Omnichain and Node Security

Different RPC URLs do not establish different operators. Agreement across stations does not establish irreversible source-chain state. Use fixed-block reads, source-chain finality, disagreement notices, replayable proofs and dispute histories. Slashing is appropriate only when errors can be publicly and objectively adjudicated, with disputes and appeals available.

17.4 Community and Research Incentive Security

Verify GitHub Webhook signatures and provenance chains for PRs, releases, CI and build identities. Prevent related-account reviews, fake citations, commit farming and appropriation of others' work. Preserve privacy and copyright boundaries. Research reviewers must disclose conflicts, and core reviewers must not use non-public information to allocate funds directly to their own projects.

17.5 Issuance and Compliance

Token sales, automated trading, referral commissions, staking rewards, buybacks, custody and return promotion may face different rules across jurisdictions. Before issuance, obtain legal advice based on the actual entity, service regions, rights and marketing, and implement appropriate territorial restrictions, abuse prevention and risk disclosures. This bluepaper is not securities, legal or investment advice.

18. Governance Separating Protocol, Treasury and Operations

We propose three governance layers: Protocol Governance for open specifications, interfaces and compatibility; Treasury Governance for budgets, grants and payment oversight; and Operations & Security for emergencies and service operations. Responsibilities must remain distinct in the permission model even if one team initially performs them.

Material upgrades, fee changes, receiving-address changes and additional powers should use multisignature authorization, timelocks, announcements and historical audit logs. Economic parameters affecting existing PoB orders should generally not apply retroactively. Emergency pauses should have predefined recovery and exit paths, rather than permit arbitrary administrator transfers.

We follow public proposal processes for standardization. Adoption of TAP-20 as a formal TapeOut standard is determined externally. Until adoption, we will use “proposed TAP-20” and disclose application status.

19. Market Positioning, Participants and Use Cases

19.1 Beyond a Conventional Launchpad

Conventional issuance infrastructure primarily optimizes asset creation and initial liquidity. TapeUP’s design combines ITO circuit delivery, a proposed native asset standard, time-distributed buy execution, public building evidence and node services in one lifecycle. Modules have different technical maturity levels; completion of one is not evidence that others are complete.

19.2 Participant Roles

Role	Core Need	Value Contributed
Project developers	Transparent launch rules and tools	Products, code, verifiable circuits
Early participants	Trading rules, exit rights, fund security	At-risk capital and market feedback
Open-source maintainers	Discovery, funding and reuse of work	Code, maintenance, security updates
Researchers	Transparent funding and replication framework	Algorithms, experiments, technical knowledge
Community contributors	Verifiable public collaboration tasks	Documentation, education, testing, operations
Metro Station operators	Measurable services and revenue	Data reading, verification, availability

19.3 Verifiable Success Criteria

Beyond public issuance, meaningful long-term measures include accepted deliverables, consistently active independent developers, reproducible research, third-party reuse, genuine paid services and sustainable ecosystem budgets after costs. PoB order counts and transaction taxes are market metrics, not substitutes for building outcomes.

20. Roadmap and Acceptance Criteria

20.1 Planned Milestones

Target Date	Milestone	Proposed Acceptance Criteria
End of October 2026	Omnichain Ledger public preview	Identities, pages and messages trace to source-chain records with confirmation status
End of November 2026	Initial TAP-20 specification, examples, application progress	Public specification, test vectors, external review status and differences
End of December 2026	PoB Trading Engine beta	Reproducible immediate/scheduled orders, Builder acceleration, point reconciliation, failure recovery
End of January 2027	Initial Metro Station tests	Inspectable circuit eligibility, task verification, contribution records and reward settlement

The table above lists our previously proposed delivery targets. Timing may change with development, security review and external standards review. The integrated ITO → automatic asset issuance path is a broader launch direction without a separate confirmed delivery date.

20.2 Proposed Product Priorities

P0: Define funds and asset architecture. Confirm ITO rights, asset interfaces, custody, fees and administrative powers; publish a read-only risk page and reviewable demonstration.
P1: Connect a complete, restricted issuance test flow. Simulate project configuration, ITO completion, asset initialization and PoB settlement end to end, including failures and refunds.
P2: Establish auditable operating data. Integrate Metro Station results, public fee accounts and provable node work.
P3: Pilot Build Ledger. Test GitHub contribution evidence, maintenance bounties, public acceptance and a small Build Treasury.
P4: Expand community and research. Explore Research Track, Builder Passport, contribution graphs and appropriate privacy proofs.

20.3 Core Evaluation Metrics

Safety and integrity: successful fund-conservation reconciliations, unexecutable order rates, refund timeliness, audit issue closure and disclosure delay for permission changes. Real adoption: independent developers, replicated functional circuits, accepted deliveries, non-farmed retention, third-party calls and actual service payments. Contribution continuity: merged and verified material PRs, maintenance completion, research replications and usage of community public goods. Operations: valid node tasks, unit verification cost, independent node count and operating profit or loss.

TVL, trading volume, token market capitalization, KOL counts and social exposure must not be the sole measures of building outcomes.

21. Risk Factors and Disclaimers

Technical risks: The protocol is under development, standards proposals are not formally adopted and core contracts may contain defects. Cross-chain reading and verification depend on public chains, RPC, upgrades and finality assumptions. Asset and market risks: Future PoB execution prices may cause losses; transaction taxes, slippage and Gas reduce receipts; liquidity may disappear; and scheduled buying may be arbitraged. Governance risks: Administrators, multisignature signers and reviewers may err, collude or abuse powers. Economic risks: Insufficient service revenue may make buybacks, grants and node rewards unsustainable; $UP rights and issuance information await formal disclosure. Legal risks: Token rights, marketing, distribution, custody, buybacks or staking may be restricted. Applicable compliance arrangements must be completed before serving specific users and jurisdictions.

Published by the TapeUP project team, this bluepaper describes product directions, existing mechanism designs and proposed implementations. It is not a public sale document, investment recommendation, price forecast, fixed-return promise or legal opinion. Users should rely on actual deployed contracts, project-specific disclosures and applicable law at issuance. We maintain this bluepaper and public disclosures, checking consistency and accuracy when releases, contract upgrades and economic parameters are finalized.

22. Conclusion Launch Begins the Commitment to Build

TapeUP seeks to connect the organizational efficiency of Launch with the continuity of Build, rather than design an attractive price curve for every project. A project may start with transistors and circuits, become an asset governed by explicit rules, attract initial market participants and then engage open-source developers, researchers and community collaborators, leaving traceable records of delivery, usage and settlement.

Make building verifiable, contributions accessible and rules reviewable, so that long-term value can emerge from actual work.

Appendix A Pre-Issuance Disclosure Status and Completeness

As of this edition, the following matters remain subject to final determination or public disclosure. This edition cannot directly serve as a public token fundraising document. Before related issuance or services open, we will provide contract-consistent details and identify inapplicable items explicitly. Plans must not substitute for completed implementation.

The table lists required disclosure categories and current status. “Not yet disclosed” does not mean an item is already determined or implemented. Formal disclosures govern all figures.

Item	Current Disclosure Status	Required Before Issuance or Service Launch
Issuer and legal entity	Not yet disclosed	Registered entity, jurisdiction, responsible party, terms
Mainnet / contract addresses	Not yet disclosed	Core addresses, Chain ID, source code, versions

Item	Current Disclosure Status	Required Before Issuance or Service Launch
ITO definition	Circuit minting and tape-out direction proposed	Participant rights, acceptance state, failure handling
Asset implementation	Proposed TAP-20	Actual TAP-20/ERC-20/wrapping approach and compatibility tests
$UP supply and circulation	Not yet disclosed	Supply, decimals, network, initial circulation, burn rules
$UP allocation and vesting	Not yet disclosed	Team, investor, ecosystem, treasury allocations and vesting
Project token allocation	Not yet disclosed	ITO/LP/team/grants, lockups, permissions
Transaction tax	Configurable from 1% to 10%	Default rate, tax base, receiving wallets, modification powers
PoB custody and refunds	Mechanism direction described	Vault control, order rights, cancellation/refund/pause
Builder Code rules	20% bilateral acceleration described	Calculation base, order, limits, anti-abuse, disputes
Buybacks and rewards	Proposed $UP allocated by valid point share	Funding, purchase method, frequency, zero-revenue treatment
Metro Station economics	Circuit staking and verification rewards described	Staked assets, settlement currency, exit/slashing/pricing
Security audits	Completion evidence not yet disclosed	Reports, remediation, test repository, emergency contacts
Governance and upgrades	Not yet disclosed	Multisignature, timelocks, permissions, changes, emergencies
Launch plan	Four indicative delivery milestones proposed	Issuance dates, trading venues, LP sources, risk disclosure

Appendix B Technical Glossary

This glossary explains TapeUP terminology and scope for issuing projects, general participants and technical reviewers. Proposed modules are not deployed protocols, and industry definitions do not establish completed TapeUP implementation.

A TapeOut and Issuance Foundations

Term	Meaning and Scope in TapeUP
TapeOut	Protocol ecosystem based on logic gates, circuits, containers and on-chain applications. TapeUP plans products on this ecosystem; this does not imply official TapeOut endorsement.
Transistor / NAND / LATCH	Transistors are basic building elements; NAND is a NOT-AND logic gate; LATCH retains state. These circuit concepts do not necessarily mean physical chip manufacturing.
Tape-out	Conventionally, submission of chip designs for fabrication. TapeUP will disclose ITO-specific on-chain construction, minting and success criteria before issuance; physical wafer fabrication is not implied.
Circuit NFT	An NFT representing a circuit asset and on-chain ownership. Ownership does not automatically establish commercial value or income rights.
Container / ERC-6551	A container is a smart contract account associated with a circuit NFT. ERC-6551 concerns NFT-bound accounts. Interfaces and powers depend on deployment.
DeWEB	TapeOut infrastructure for decentralized website access and content. Distinguish on-chain content verification from frontend gateway convenience.
ITO (Initial Tape-Out)	TapeUP’s term for transistor minting and circuit construction/minting leading to launch conditions. Participation rights and completion criteria await final determination and disclosure.
TAP / TAP-10	TAP is TapeOut’s technical proposal process. TAP-10 concerns DeWEB access and messaging. Draft proposals are not binding final standards.
Proposed TAP-20	TapeUP’s application name for a circuit/container-native asset specification. As of the reference date, the related PR was closed and unmerged; the proposal was not adopted as a formal standard.
ERC-20 / EVM	ERC-20 is a common fungible-token interface; EVM is the Ethereum Virtual Machine execution environment. Wallet and DEX support depends on actual interfaces.
Launch Factory / ITO Gateway	Proposed TapeUP issuance-orchestration contracts and ITO state integration modules, not claims of deployed code.
TokenLog / Runtime / AssetAdapter	Proposed operation logs, execution module determining valid asset state, and interface compatibility layer. Final authority and implementation remain to be determined.
Wrapped Token	Locking an underlying asset and minting a compatible trading token. Locked assets and wrapped supply must conserve value, with redemption and custody risks addressed.

B PoB and Market Mechanisms

Term	Meaning and Scope in TapeUP
PoB (Proof of Build)	A philosophy emphasizing verifiable long-term delivery, usage and contributions, distinct from the specific scheduled-purchase engine.
PoB Trading Engine	TapeUP's proposed scheduled-purchase, Builder Code and points mechanism. It changes buyer-fund timing, not guaranteed price direction.
AMM	Automated market maker: swaps through liquidity pools and pricing functions. Prices depend on liquidity, trading and arbitrage.
DEX	Decentralized exchange using contracts for swaps. Assets must satisfy relevant interfaces and routing requirements.
LP / Liquidity	LP means liquidity provider or liquidity share; liquidity means executable market depth. Issuance does not automatically provide adequate liquidity.
Slippage	Difference between expected and actual execution price; potentially large in thin or volatile markets.
Gas	Network fees for on-chain execution, which can make small scheduled purchases uneconomic.
MEV	Extractable value from ordering, inserting or front-running transactions. Predictable scheduled buying requires execution safeguards.
Vault	Contract holding unspent user purchase budgets under order rules. Ownership, authorized uses and refunds must be explicit.
Scheduler	Module determining when orders become executable, who triggers them and how failures retry. It does not ensure successful execution of every order.
Builder Code	Collaboration identifier triggering bilateral purchase acceleration in the PoB design. It is not proof of actual developer contributions.
Buying Reservoir / Scheduled Budget	Unfilled order funds reserved for future purchases. These are liabilities or restricted funds, not protocol revenue or completed buying.
Nullifier	Cryptographic identifier preventing repeated use of one qualification. Alone, it generally cannot prevent one person from creating multiple qualified identities.
Buyback	Protocol purchase of a specified token using funds permitted by disclosed rules. It is not continuous price support or a return guarantee.
Fee Splitter / Transaction Tax	Allocation of received fees or transaction taxes under explicit rules. Tax base, sequence, rate and beneficiary addresses must be disclosed.
Tokenomics	Supply, allocation, circulation, vesting, functions and fund flows. A single tax rate or reward algorithm cannot substitute for a complete model.
$UP	Identifier for TapeUP's proposed PoB buyback-reward token. Issuance status, supply, contracts, rights and allocation await formal disclosure.

C Development, Research and Community Building

Term	Meaning and Scope in TapeUP
Build Ledger	Proposed traceable index of repositories, commits, releases, circuits, acceptance and settlement, not a simple GitHub activity leaderboard.
Build Treasury	Proposed dedicated funding pool for development, research, maintenance and community work under public budgets, separate from user PoB principal.
Grant / Bounty	Grants generally support a research or building process; bounties reward specific tasks or outcomes. Acceptance criteria must be announced beforehand.
Build-to-Unlock	Optional milestone-based release of designated building budgets. It must not arbitrarily restrict ordinary user withdrawals.
Builder Passport	Proposed record linking verifiable contribution history, not mandatory real-name identification or an irreversible, unappealable reputation score.
Contribution Graph	Proposed cross-project citation and collaboration graph for reuse and attribution. Citations alone must not automatically determine payouts.
GitHub / Repository	GitHub hosts code collaboration; a repository stores code. Activity does not directly prove security or commercial adoption.
Commit / Commit SHA	A commit records a version change; its SHA identifies specific contents. It locates a version but does not measure contribution value.
PR / Issue	Pull requests propose code changes and review; issues record problems or tasks. A merged PR is not necessarily high quality.
CI / Release	Continuous integration automatically tests code; a release is a published version. Reliable outcomes generally require actual operation, replication and third-party acceptance.
Webhook	Automated event notification from an external platform. Receivers must verify signatures, prevent replay and retain provenance.
Research Track / Replication	Research and replication work focused on methods, data, reproducible experiments and knowledge value, not rankings by paper count.
DOI / Benchmark	DOI is a persistent digital object identifier; a benchmark is a comparable evaluation requiring environment, inputs and repetition methods.

D Nodes, Security, Governance and Verification

Term	Meaning and Scope in TapeUP
Metro Station	A planned TapeUP node for multichain reading, data verification and service tasks, not a physical transport facility.
Staking / Slashing	Staking supplies eligibility or economic constraints. Slashing is a preagreed penalty requiring objective evidence, appeals and explicit execution rules.
SLA	Service Level Agreement specifying response time, availability and service quality measures.
RPC / Indexer	RPC interfaces query chain state; indexers organize data for retrieval. Query convenience does not replace authentic source-chain state.
Finality / Reorg	Finality means reversal risk falls below an agreed threshold; reorg means chain reorganization. Confirmation policies differ by chain.
Hash / configHash / Nonce	Hashes fingerprint data; configHash binds a configuration version; nonces order operations and prevent signature replay.
On-chain / Off-chain	On-chain means chain-recorded execution; off-chain means external computation or storage. Off-chain evidence requires explicit verification methods.
Multi-sig / Timelock	Multisignature requires joint authorization; a timelock delays sensitive actions for public observation.
ZK	Zero-knowledge proofs establish computational relationships without exposing all private inputs. They do not automatically establish external input truth.
APR / TVL / KPI	Annual percentage rate, total value locked and key performance indicators. None establishes long-term value or guaranteed returns.
Sybil Attack	One controller creates multiple identities to duplicate eligibility or manipulate evaluation. Wallet count does not equal person count.
Audit / Independent Review	Systematic review of code, security or finances. A passed audit may be claimed only after independent completion with scope and findings disclosed.
Governance / Protocol Upgrade	Decision and authorization processes. Upgrades may change contract logic; permissions, procedures and effects on existing rights must be disclosed.

Appendix C Sources and Technical Basis

This bluepaper draws on TapeUP's product designs and mechanism documents, with external proposals informing standards status and verification boundaries. External technical status was checked as of October 11, 2026.

- **TapeUP team, "TapeUP Bluepaper (Draft)" (Google Docs):** foundation for the product sequence, planned milestones and conceptual positioning.
- **TapeUP team, "PoB Trading Engine:** Turning Buying Into Sustained Building" (four-page PDF): trading parameters, Builder Code and $UP buyback rewards.
- **TAP-20 proposal Issue #21 and PR #23:** the PR was closed and not merged as of the reference date; this bluepaper therefore uses "proposed TAP-20."

- **TAP-10 Draft:** technical assumptions for DeWEB access and source-chain reading.

The long-term development, research and community framework is a proposed TapeUP extension requiring development, testing and applicable governance. It does not imply deployed modules.

Appendix D Reproducible Assumptions for the PoB Experiment

This experiment explains parameter-driven differences in capital timing, not actual project prices. It compares equal 100-unit purchase budgets over 180 days under a common external disturbance sequence. Transaction taxes, fees, liquidity feedback, slippage, strategic selling, failed execution, automatic acceleration and actual AMM reserves are excluded.

Item	Assumption
Observation period	Issuance day t=0 through t=180; 181 observations
Models	Immediate 100%; 60% + 14 days; 30% + 25 days; 20% + 90 days
Total budget	100 units per model; all execution assumed successful
Price proxy	I(t)=exp(Σ[g_s+ε_s+0.0015×buy(s)])
Shared disturbance	Fixed random seed 139; daily standard deviation 0.003; disturbance at t=0 set to zero
Stress scenario g	t=0-14: −0.012/day; 15-64: −0.0032/day; 65-180: −0.001/day
Recovery scenario g	t=0-15: −0.008/day; 16-104: +0.003/day; 105-180: +0.0006/day

A deliberately linear log-impact model isolates the mathematical effect of budget scheduling. Real markets do not follow these assumptions. The charts must not be used to infer lower drawdown probability, price protection or safer parameter tiers.

After launch, we propose testing the model and mechanism against real scheduled and executed orders, cancellations and refunds, average slippage, Gas costs, DEX depth, actual participant counts, developer delivery and real service revenue.

Reproducible implementation: The accompanying Python simulation script, pob_scenario_model.py, can regenerate the charts.